

Exam Choice

Student Number

2023

TRIAL
EXAMINATION

Physics

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black pen.
- Draw diagrams using pencil.
- For questions in Section II, show all relevant working in questions involving calculations.
- NESA approved calculators may be used.

Total marks: 100

Section I – 20 marks (pages 3 – 13)

- Attempt questions 1 – 20.
- Allow about 35 minutes for this section.

Section II – 80 marks (pages 14 – 29)

- Attempt questions 21 – 36
- Allow about 2 hours and 25 minutes for this section.

Section I – 20 marks

Attempt Questions 1-20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A  B  C  D 

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

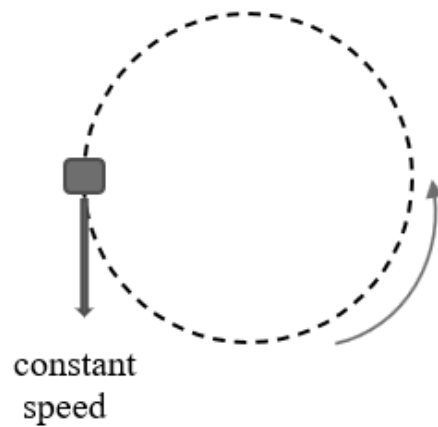
A  B  C  D 

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A  B  C  D 

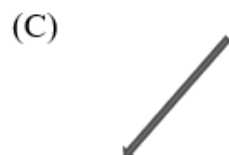
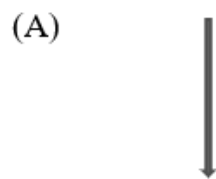
correct 

1. An object is undergoing circular motion as shown in the diagram.



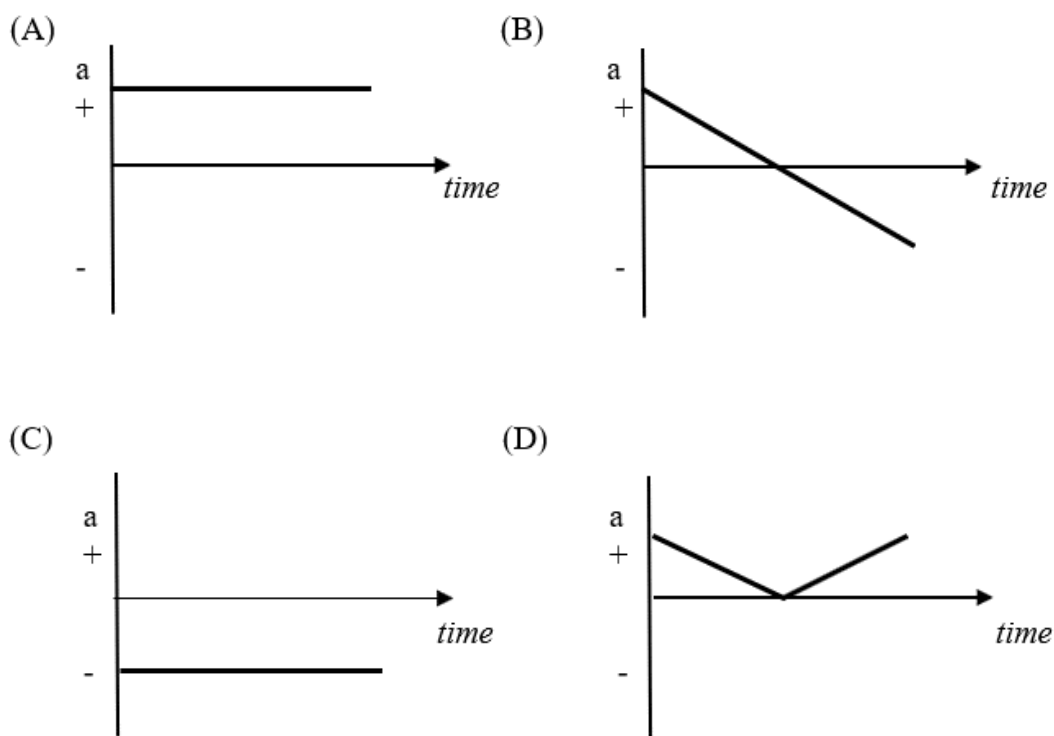
At the point shown, the object's speed is constant.

Which vector below could represent the object's acceleration?



2. A projectile is launched at 30° to the horizontal.

Which alternative below represents the projectile's acceleration while it remains in flight, assuming the upwards direction is positive?



3. An investigation was performed using a laser and a slide with two narrow slits spaced apart a small distance.

The laser light that passed through the slits was projected onto a dark wall. The resulting pattern is shown below.

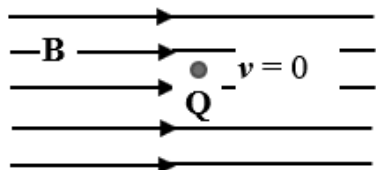


To produce a similar pattern, the light used must be:

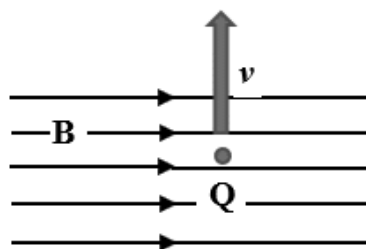
- (A) monochromatic.
- (B) polarised.
- (C) intense.
- (D) longitudinal

4. A charged particle Q is moving with velocity $\mathbf{v}$ through a magnetic field $\mathbf{B}$.

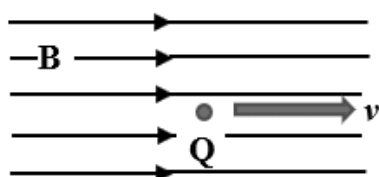
(i)



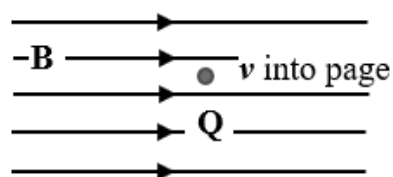
(ii)



(iii)



(iv)



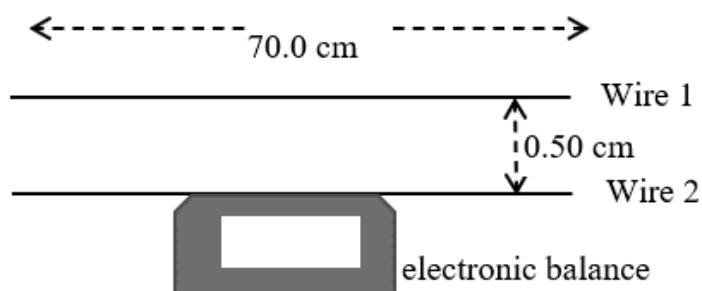
In which situations would Q experience a force due to the presence of magnetic field $\mathbf{B}$?

- (A) i and ii only
- (B) ii and iii only
- (C) ii and iv only
- (D) all situations

Questions 5 and 6 refer to the following information

An investigation was performed with the aim of measuring the value of the magnetic permeability constant, μ_0 .

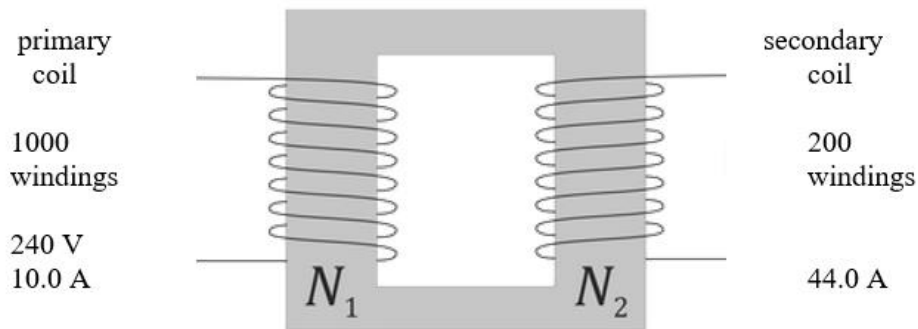
Two wires, both of length 70.0 cm were arranged horizontally, with one wire above the other, as shown in the diagram below. The bottom wire is resting on an electronic balance.



When a current of 25.5 A flows through both wires, the reading on the electronic balance changes by 0.500 g.

5. What is the value of μ_0 as determined by this investigation?
- (A) 3.4×10^{-7}
 - (B) 1.7×10^{-9}
 - (C) 3.4×10^{-9}
 - (D) 1.3×10^{-7}
6. Which measurement made in the above investigation has the greatest relative error?
- (A) length of the wires
 - (B) separation of the wires
 - (C) current in wires
 - (D) change in the reading of the electronic balance

7. A transformer has 1000 windings on the primary coil and 200 windings on the secondary coil.



The primary voltage is 240 V, with a primary current of 10.0 A.

The secondary current is 44.0 A.

What is the efficiency and type of this transformer?

	Efficiency	Type
(A)	23%	step-up
(B)	44%	step-up
(C)	20%	step-down
(D)	88%	step-down

8. A radioactive sample's level of radioactivity was found to follow the formula:

$$N_t = N_0 e^{-0.024t} \quad \text{where } t \text{ is in hours.}$$

The half-life of this radioactive sample is closest to:

- (A) 29 hours
(B) 0.024 hours
(C) 0.024 seconds
(D) 1/0.024 hours

9. By repeating an investigation without changing the procedure, what can be tested?

- (A) reliability
- (B) validity
- (C) accuracy
- (D) percentage error

10. The spectrum below was obtained using hydrogen.

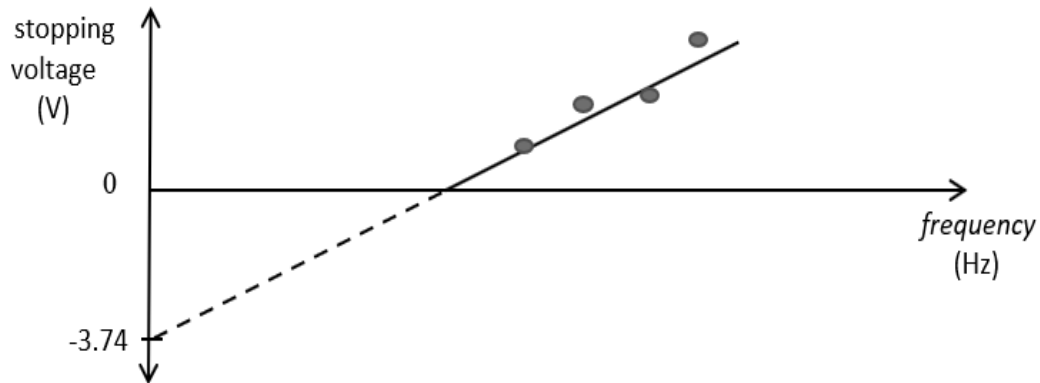


A similar spectrum could be obtained by observing light from a:

- (A) very hot object.
 - (B) star.
 - (C) gas discharge tube.
 - (D) fluorescent lamp.
11. For a planet orbiting a single star, which factors determine the planet's orbital period?
- (A) mass of planet; mass of star; radius of orbit
 - (B) mass of star; radius of orbit
 - (C) speed of planet; radius of orbit
 - (D) mass of star; mass of planet

12. An investigation was performed into the photoelectric effect using a zinc cathode.

The results were plotted on a set of axes as shown below and extrapolated to the y-axis where the line of best fit intersects at -3.74 V.



Which statement below is correct?

- (A) 3.74 V of potential difference is required to be applied to the cathode for electrons to be released.
 - (B) The threshold frequency is close to 9.0×10^{14} Hz.
 - (C) More photoelectrons are emitted as the frequency of the incident light increases.
 - (D) Replacing the zinc cathode with a copper cathode will not change the threshold frequency.
13. The Standard Model of matter states that protons are not considered to be fundamental particles because:
- (A) protons may spontaneously decay into other particles.
 - (B) fundamental particles have no charge.
 - (C) protons can be broken apart in particle accelerators.
 - (D) protons are a member of the Boson family of particles.

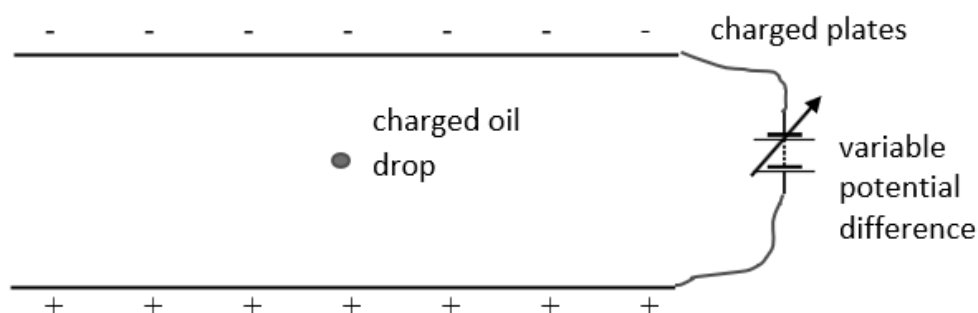
14. Of the discoveries listed, the one best linked to the discovery of the expansion of the Universe is:

- (A) the application of the Doppler effect to light from other galaxies.
- (B) the discovery of gas and dust clouds in the galaxy.
- (C) astronomers realising that stars have a variety of colours.
- (D) computer simulations showing galaxies are moving through space.

15. An investigation was performed to simulate Millikan's oil drop experiment.

The apparatus was arranged such that small, charged oil drops were placed between charged plates.

Each oil drop has a mass of $5.0\ \mu\text{g}$.



It was observed that when the potential difference was set to certain values that some oil drops would be suspended between the charged plates.

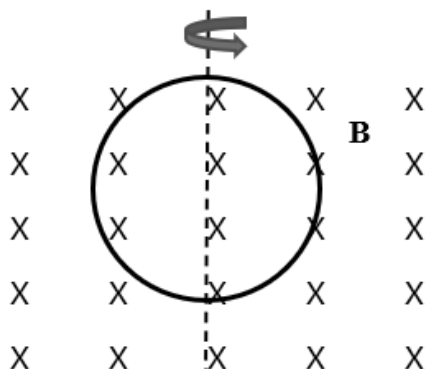
Similar observations allowed Millikan to calculate the charge on individual electrons.

What did Millikan use to achieve his results?

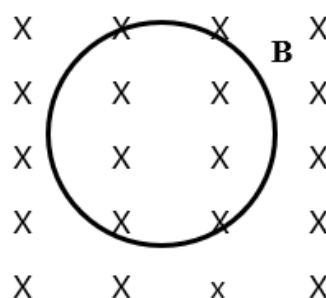
- (A) The mass of the oil drops he was using.
- (B) The speed of the oil drops that were not suspended.
- (C) The difference between the potential differences that caused some of the oil drops to be suspended.
- (D) The radius of the oil drops that were suspended.

16. In the diagrams below the loops are initially all the same size and the magnetic field B is the same.

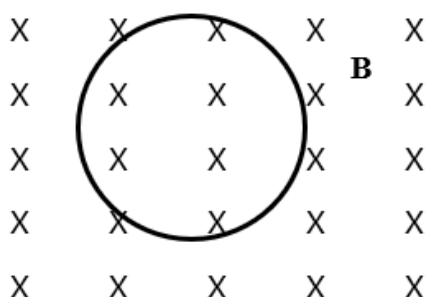
(S) The loop is rotated 90° in 2.0 s.



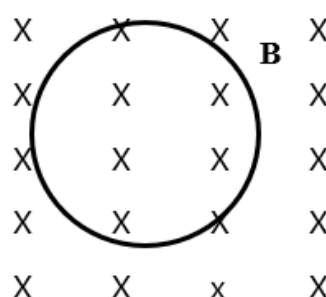
(T) The magnetic field is reduced to zero in 4.0 s.



(U) The loop's radius doubles in 2.0 s.



(V) The direction of the magnetic field is reversed in 2.0 s.



In which situation would the average induced EMF around the loop be the greatest?

- (A) S
(B) T
(C) U
(D) V

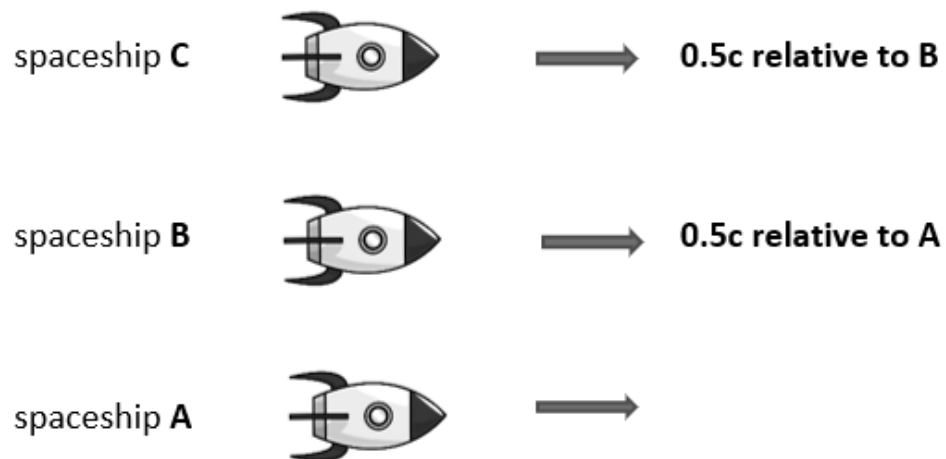
- 17.** A limitation of Bohr's model of the atom is:
- (A) it does not account for the nucleus.
 - (B) it cannot explain the absorption spectrum of hydrogen.
 - (C) only the first few electron energy level transitions in hydrogen are explained.
 - (D) it does not predict the spectrum for multi-electron atoms.
- 18.** The process of nuclear fission is used in nuclear power stations in many countries around the world to produce electricity.
- One reason why nuclear fusion is not also used is:
- (A) fusion reactions do not release sufficient energy to produce electricity.
 - (B) there is an insufficient supply of fuel for nuclear fusion on Earth.
 - (C) the products of nuclear fusion are highly radioactive.
 - (D) nuclear fusion is very difficult to start.
- 19.** Which statement relating to AC induction motors is correct?
- (A) There is no current produced in induction motors.
 - (B) Induction motors need slip rings to operate.
 - (C) The magnetic field rotates faster than the rotor.
 - (D) The rotor rotates faster than the magnetic field.

20. Three identical spaceships are moving through space in the same direction.

The spaceships are each **100 m** long when measured at rest.

Observers onboard spaceship **A** measure spaceship **B**'s speed relative to them as being $0.5c$.

Observers on spaceship **B** measure spaceship **C**'s speed relative to them as being $0.5c$, as shown.



If an observer on spaceship **A** measured the length of spaceship **C** the result would be closest to:

- (A) greater than 100 m
- (B) 100 m
- (C) 87 m
- (D) less than 87 m

2023

TRIAL
EXAMINATION

Physics

Section II Answer Booklet

80 marks

Attempt Questions 21 – 36

Allow about 2 hours and 25 minutes for this part

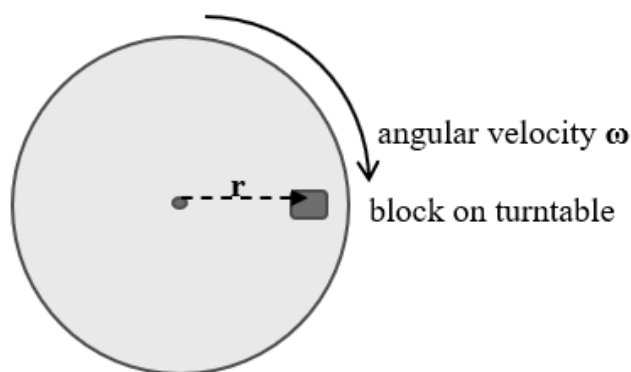
Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Show all relevant working in questions involving calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which questions you are answering.

Question 21 (5 marks)

An investigation was performed to measure the frictional force between a block and the surface of a turntable.

The block, mass m , was placed a distance r from the centre of the horizontal turntable, which was turned with an increasing angular velocity ω until the block just began to slide, as shown.



- (a) Show how the centripetal acceleration of the block could be calculated using the values of r and ω .

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- (b) Explain why the block eventually begins to slide on the turntable once the angular velocity reaches a certain value.

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Question 22 (8 marks)

Kepler published his Laws of Planetary Motion before Newton's Law of Universal Gravitation.

- (a) Discuss how Kepler's 2nd Law relates to Newton's Law of Universal Gravitation. 2

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- (b) i. Given that Earth is in orbit about the Sun, which has a mass of 2.0×10^{30} kg at an average distance of 150 million km, what is Earth's gravitational potential energy? 2

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- ii. Explain why the value of Earth's potential energy is negative. 1

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- (c) What speed would the Earth need to attain so that it could escape the gravitational field of the Sun? 3

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Question 23 (4 marks)

Jupiter is about 26,000 times more massive than the Moon.

Using appropriate equations, discuss how the gravitational effect of Jupiter has a negligible effect on Earth's tides compared to the Moon.

4

You may use the following data:

Mass of the Moon = 7.3×10^{22} kg

Closest distance between Earth and Moon = 3.6×10^8 m

Mass of Jupiter = 1.9×10^{27} kg

Closest distance between Earth and Jupiter = 5.9×10^{11} m

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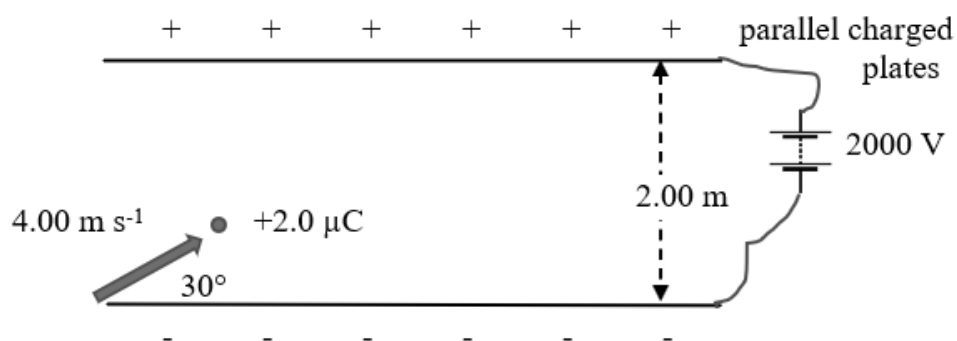
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Question 24 (6 marks)

A small particle with mass 5.0 mg and a charge of $+2.0 \text{ } \mu\text{C}$ is shot at an angle of 30° and initial speed 4.00 m s^{-1} into a large gravity-free region between two parallel charged plates 2.0 m apart, as shown below.



- (a) Find the acceleration of the charged particle within the gravity-free region.

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- (b) How far along from its launch position will the charged particle hit the negative plate?

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Question 25 (4 marks)

Using relevant equations, describe how the stopping voltage applied to electrons emitted from the cathode of a photoelectric effect apparatus enables us to find the maximum kinetic energy of these emitted electrons.

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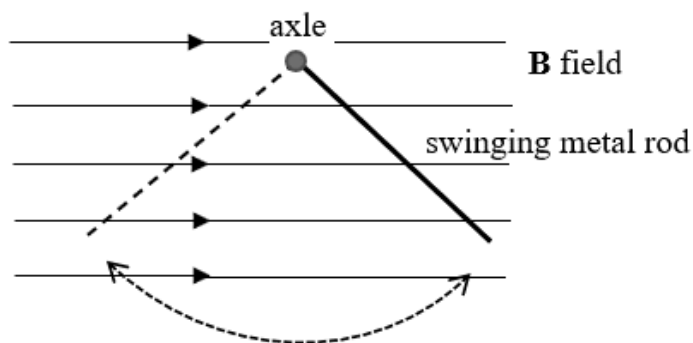
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Question 26 (5 marks)

A metal rod swings freely back and forth around an axle within a uniform magnetic field, as shown.



- (a) When the metal rod is mid-way through its swing, it is vertical.

Explain why, at this instant, as well as when it is at either end of its swing, the EMF induced across the ends of the rod is zero.

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- (b) When the metal rod is stationary at its vertical position, a current is made to flow through it from top to bottom.

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Explain whether or not this will cause the rod to swing.

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Question 27 (8 marks)

Step-up and step-down transformers allow for the distribution of electrical energy from power stations to consumers in an efficient manner.

Analyse the applications of such transformers and describe techniques used to limit the loss of power in the transformers as well as in the transmission wires.

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Question 28 (4 marks)

James Clerk Maxwell's contributions to classical physics were made through his description of electromagnetism in mathematical formulae.

Outline the major aspects of Maxwell's contributions to electromagnetic theory at the time.

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Question 29 (2 marks)

A person lost overboard from a cruise ship can be seen at night using night vision cameras.

Assuming the person has a body temperature of 37°C, what is the peak wavelength of emitted radiation from the person?

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Question 30 (4 marks)

The light transmitted from some aircraft instrument displays is horizontally polarised.

- (a) Explain why aircraft pilots avoid wearing polarised sunglasses. 2

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- (b) To reduce the transmitted light through a window, 2 layers of polarised glass are used.
- If 30% of the original light is to be transmitted, how should the two layers of glass be oriented to each other? 2

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Question 31 (5 marks)

Einstein postulated that all inertial frames of reference are equivalent.

- (a) Describe what Einstein meant by an “inertial” frame of reference. **2**

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- (b) Using an example, describe how observations of the same event made from two different inertial frames of reference can be different. **3**

A diagram should be used to illustrate your answer.

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Question 32 (4 marks)

As a spacecraft approaches the speed of light, the extra kinetic energy provided by the thrust from its propulsion system results in very little increase in speed.

With reference to appropriate mathematical relationships, explain this apparent discrepancy in Newtonian physics theory in terms of special relativity. Refer to any relevant formulae.

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Question 33 (6 marks)

Nucleosynthesis that occurs in main sequence stars can be summarised as the conversion of hydrogen nuclei to helium nuclei with the consequential emission of energy.

- (a) Outline how this process is able to emit large amounts of energy. **2**

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- (b) Given that the probability of collisions involving more than two particles is very small, outline the pathways believed to be followed in the nucleosynthesis of helium in main sequence stars. **4**

Your response does not need to include the details of every step in the processes.

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Question 34 (5 marks)

Rutherford's atomic model placed all of the positive charge and most of the mass of an atom in the nucleus. It could not account for the existence of isotopes, where atoms of the same element differ in mass.

Outline the way in which the discovery of the particle responsible for this variation in mass was made, and explain why the discovery came so much later.

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Question 35 (7 marks)

- (a) Using an example of a proton moving at 500 km s^{-1} , show that its wavelength is too short to detect using present-day technology. **3**

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- (b) Outline the experimental evidence that led to the development of the formula used to calculate the wavelength of de Broglie's matter waves. **4**

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Question 36 (3 marks)

A technique used to turn pure silicon-28 into a type of material used in semiconductors is to bombard the pure silicon-28 with neutrons from a nuclear reactor.

Using appropriate equations, show how the capture of a neutron by a silicon-28 nucleus can result in the transmutation of silicon nuclei into phosphorous by beta decay.

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Section II extra writing space.

If you use this space indicate clearly which question you are answering.

[illegible]

Section II extra writing space.

If you use this space indicate clearly which question you are answering.

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Exam Choice

2023 Physics Trial Examination.

Marking Guidelines and Model Answers.

Section I Multiple Choice

1	2	3	4	5	6	7	8	9	1	1	1	1	1	1	1	1	1	1	2
									0	1	2	3	4	5	6	7	8	9	0
D	C	A	C	A	B	D	A	A	B	B	B	C	A	C	D	D	D	C	D

Section II

21 a.

Marking Criteria	Marks
• complete explanation provided	2
• incomplete/partial explanation provided	1

Use $\omega = \Delta\theta/t$ with $\theta = 2\pi$ rad to find T.
Then use $v = 2\pi r/T$ and finally, use $a_c = v^2/r$

21 b.

Marking Criteria	Marks
• Full explanation with correct working shown	3
• Partial explanation provided	2
• A correct aspect/fact provided	1

The increasing angular velocity ω results in increasing centripetal acceleration, a_c . As $F_c = ma_c$, the force required to maintain the block's circular motion also increases until it is just greater than the friction between the block and the surface of the turntable, at which point the block begins to slide towards the outer edge of the turntable.

22 a.

Marking Criteria	Marks
• Kepler's 2 nd Law identified	2
• Law is linked to motion under gravity and the nature of gravity and distance	
• Kepler's 2 nd Law is identified	1

Kepler's 2nd Law of Planetary Motion states that a planet will sweep out equal areas in equal time in its orbit about the Sun. This requires faster motion of the planet as it moves closer to the Sun, which occurs due to the inverse square law nature of gravity, which increases as the planet moves closer and loses gravitational potential energy, gaining KE.

22 b. (i)

Marking Criteria	Marks
• Calculations are clear and correct	2
• Calculations made but with an error	1

$$U = -G \frac{Mm}{r}$$

$$= - (6.67 \times 10^{-11}) \times (2.0 \times 10^{30}) \times (6.0 \times 10^{24}) / (150 \times 10^9)$$

$$= - 5.3 \times 10^{33} \text{ J}$$

22 b. (ii)

Marking Criteria	Marks
Explanation related to the definition of U provided	1

To provide a working universal definition for U, it is taken that beyond a gravitational field, $U=0$ and as a body falls towards another mass, it loses U so that the value is negative.

22 c.

Marking Criteria	Marks
- Correct answer with working provided using value obtained in Q22b(i) - Initial assumption $U+K=0$ made	3
- Working provided but an error made - Initial assumption $U+K=0$ made	2
A correct step towards the answer provided	1

$$U + K = 0$$

$$K = 5.3 \times 10^{33} \text{ (from Q22b(i))}$$

$$\frac{1}{2}mv^2 = 5.3 \times 10^{33}$$

$$v = \sqrt{\frac{5.3 \times 10^{33}}{0.5 \times 6.0 \times 10^{24}}}$$

$$v = 4.2 \times 10^4 \text{ m s}^{-1}$$

23

Marking Criteria	Marks
- Discussion answers question posed - Discussion backed by calculations which are referred to - Response is logical and concise	4
- Discussion addresses question posed - Some attempt at calculations made	3
Several aspects of the question addressed	2
A relevant aspect of the question is addressed	1

As gravity obeys the inverse square law, distance between masses involved is very important. Although Jupiter is thousands of times more massive than the Moon, its distance away reduces its gravitational force on the Earth to only a small fraction that of the Moon.

Using $F = \frac{GMm}{r^2}$ for Jupiter-Earth and then for Moon-Earth:

$$F_{\text{Jupiter-Earth}} = (6.67 \times 10^{-11}) \times (6.0 \times 10^{24}) \times (1.9 \times 10^{27}) / (5.9 \times 10^{11})^2$$

$$= 2.2 \times 10^{18} \text{ N}$$

$$F_{\text{Moon-Earth}} = (6.67 \times 10^{-11}) \times (6.0 \times 10^{24}) \times (7.3 \times 10^{22}) / (3.6 \times 10^8)^2$$

$$= 2.3 \times 10^{20} \text{ N}$$

As shown, even at its closest approach to Earth, Jupiter exerts around 1% of the gravitational force as the Moon, so that its effect on Earth's tides is negligible.

24 a.

Marking Criteria	Marks
Correct steps taken with correct working to obtain correct answer.	3
As above but with an error	2
A correct step provided	1

Use: $F = qE$

$$F = qV/d \text{ and as } F = ma,$$

$$ma = qV/d \text{ so}$$

$$a = qV/md$$

$$= (2.0 \times 10^{-6}) \times (2000) / ((5.0 \times 10^{-6}) \times (2.00))$$

$$= 4.0 \times 10^2 \text{ m s}^{-2} \text{ downwards towards the negative plate}$$

24 b.

Marking Criteria	Marks
• Correct method used to find correct answer (carry-over error OK)	3
• Correct method attempted with error made	2
• A correct step completed towards obtaining an answer	1

$$\begin{array}{lll}
 u_y = u \sin \theta & u_x = u \cos \theta & \text{Use: } v = u + at \\
 u_y = 4.00 \sin 30^\circ & u_x = 4.00 \cos 30^\circ & -2.00 = 2.00 + -4.0 \times 10^2 t \\
 = 2.00 \text{ m s}^{-1} & = 3.46 \text{ m s}^{-1} & 400t = 4 \\
 & & t = 0.01 \text{ s}
 \end{array}$$

$$\begin{aligned}
 \text{So: } \Delta x &= u_x t \\
 &= 3.46 \times 0.01 \\
 &= 3.46 \times 10^{-2} \text{ m}
 \end{aligned}$$

25

Marking Criteria	Marks
• Response is detailed and comprehensive, includes relevant equations and concepts	4
• Response addresses the concepts in the question	3
• Several aspects of the question are addressed	2
• A relevant aspect of the question is addressed	1

When electrons are emitted from the surface of the cathode in the photoelectric effect they have kinetic energy which is the remaining energy from the absorbed photon of incident light. When experiencing an electric field in a direction that opposes the motion of the electron, work is done: $W = qV$ on the electron and the electron slows down. If the potential difference V of the field is adjusted so that the electron stops just short of the anode, the photo current drops to zero. Knowing the charge on an electron and this “stopping voltage” allows us to calculate the maximum kinetic energy of the emitted electrons.

26 a.

Marking Criteria	Marks
• Explanation for reasons at both positions is clear and correct	3
• Response is coherent and concise	2
• Explanation for either position is made thoroughly	1
• A partial explanation for either position is provided	

When vertical, the electrons/charges in the metal rod are moving parallel to the direction of the magnetic field. As $F = qvB \sin \theta$ and here, $\theta = 0^\circ$ or 180° so no force acts on the charges in the rod, so no potential difference is induced across the ends of the rod. At either end of the rod's swinging motion, $v = 0$, so again, no force acts on the charges in the rod and no induced potential difference is induced.

26 b.

Marking Criteria	Marks
• Correct answer with correct reasoning given clearly	2
• Correct answer but reasoning unclear or insufficient	1

The force on the rod due to the current flowing down the rod within the magnetic field is out of the page, not in the plane of the rod's swinging motion, so the force resulting from the current will not cause the rod to move in its plane of motion.

27

Marking Criteria	Marks
- A comprehensive and thorough analysis provided - Response includes reference to both step-up and step-down transformers - At least two methods used to ensure the efficiency of transformers are described clearly - Response is logical and coherent and uses appropriate physics terminology and refers to relevant physics principles	8
- An analysis provided - Response includes reference to step-up and step-down applications of transformers - At least one method used to ensure transformer efficiency is described - Appropriate physics terminology used	6-7
- Some analysis provided - Reference to the application of transformers provided	4-5
Response includes reference to applications of transformers or a method to ensure their efficiency	2-3
Response includes at least one relevant reference	1

Power lost in the transmission of electrical energy through wires often hundreds of kilometres long, connecting power stations to the end consumers, is proportional to the square of the current flowing. Given $P=VI$, a higher voltage allows for a lower current for the same power transmitted. Hence step-up transformers are used very close to the source of generation, usually boosting the voltage to 330 or 500 kV for long distance transmission. Such voltages can arc many metres through air and are thus extremely dangerous, so step-down transformers are used to allow for the safer transmission of electrical energy in local suburbs, with further step-down transformers used to reduce the voltage to that used in households.

Transformers are constructed using a soft iron core due to its high magnetic permeability, which minimises flux leakage from the primary to the secondary coil. As $\epsilon = -N \frac{d\Phi}{dt}$, any flux leakage will reduce the EMF induced in the secondary coil and reduce the transformer's efficiency. Further, the core is laminated to reduce the size of eddy currents which are generated in the core, producing heat and therefore energy loss. Smaller eddy currents reduce this loss.

28

Marking Criteria	Marks
- Thorough outline provided referencing at least three specific contributions - Response is clear, concise and logical	4
An outline provided referencing at least two contributions in detail	3
Several appropriate and relevant pieces of information provided	2
An aspect of Maxwell's contribution is identified	1

Maxwell's mathematical equations relating to electromagnetic radiation contributed several aspects to the understanding of its nature, including:

- * the prediction of other forms of EMR other than light (which were subsequently discovered, including UV, IR, radio etc);
- * that all EMR was self-propagating, i.e. could travel through a vacuum;
- * EMR travelled at about $3 \times 10^8 \text{ m s}^{-1}$ through a vacuum (the speed of light);
- * EMR is oscillating perpendicular electric and magnetic fields.

29

Marking Criteria	Marks
Correct solution given using correct working	2
Solution obtained but with an error	1

Using Wien's Law: $\lambda_{\max} = \frac{b}{T}$ where $T = 37 + 273 = 310 \text{ K}$
 so: $\lambda_{\max} = 2.898 \times 10^{-3} / 310$
 $= 9.3 \times 10^{-6} \text{ m}$

30 a.

Marking Criteria	Marks
An appropriate explanation is given referencing the nature of polarised light and its transmission	2
A partial or incomplete explanation is provided	1

The pilot's polarising sunglasses would block the transmission of the light from the plane's instruments, as polarised sunglasses are vertically polarised giving a value of θ of 90° where $I = I_{max}\cos^2\theta$ for transmitted light. This would totally block the instrument's display.

30 b.

Marking Criteria	Marks
Correct solution provided with correct working	2
Attempt at solution made using appropriate formula	1

Using $I = I_{max}\cos^2\theta$ with $I/I_{max} = 0.30$

$$0.30 = \cos^2\theta$$

$$\cos\theta = \sqrt{0.30}$$

$$\theta = 57^\circ$$

So the two polarising layers of glass should be oriented at 57° to each other so that 30% of the incident light is transmitted.

31 a.

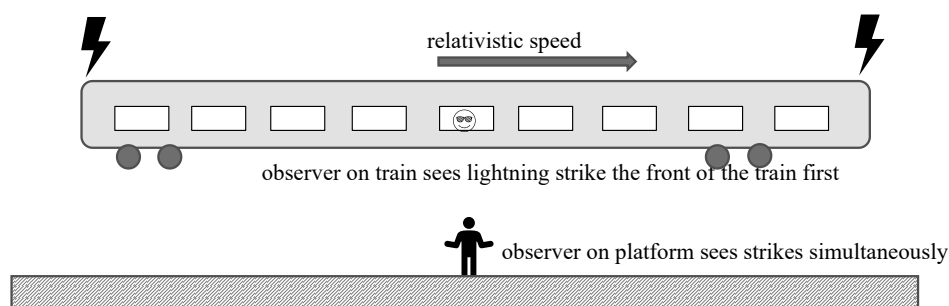
Marking Criteria	Marks
Clear, concise and correct description provided	2
Response addresses equality and non-accelerating nature of FoR	1
Partial and incomplete description provided	1

An inertial frame of reference is one where the laws of Physics hold true and no "inertial" forces need to be applied to explain observations made. Inertial FoR are non-accelerating and there is no superior inertial FoR as all are considered equivalent by Einstein's relativity.

31 b.

Marking Criteria	Marks
- A full and complete description is provided using a clearly stated relevant example - Diagram is provided and referred to	3
- An incomplete description is provided - A diagram is evident	2
An aspect of the question is correctly addressed	1

Lightning is observed striking the front and rear of a train occurring simultaneously by an observer on the platform as the train passes. The train is moving with constant velocity at relativistic speed. A passenger on the train sitting midway between front and back will observe the lightning strike the front of the train before the rear, different to the other observer. However, both observers are in inertial frames of reference as they are not accelerating, so both observations are said to be valid without one being superior to the other.



32

Marking Criteria	Marks
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• Response includes explanation of mass dilation • Reference to KE and mass dilation equations made in explanation • Response is clear, logical and concise	4
• Explanation of mass dilation provided • Reference to one equation made	3
• Partial description of mass dilation provided	2
• One relevant aspect of mass dilation identified	1

Work is still being done on the spaceship as it approaches the speed of light, $W = F \times s$, thus there is still an increase in kinetic energy, $KE = 1/2mv^2$. As v is not increasing, the only other quantity that can increase is m . Einstein considered this situation as an aspect of his special relativity. Known as mass dilation, this increase in mass is given by: $\rho_v = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}$ noting that p represents momentum, $p = mv$.

33 a.

Marking Criteria	Marks
• Response includes a detailed outline of the origins of the energy stars produce	2
• Some aspect of fusion is provided	1

This process is known as the fusion of hydrogen to helium. The mass of the products is less than that of the reactants. The mass difference between the two is converted into energy by the formula $E = mc^2$. Only a small amount of mass difference can result in large amounts of energy being produced. In stars, so much fusion is happening that it is sufficient to keep the star from collapsing.

33 b.

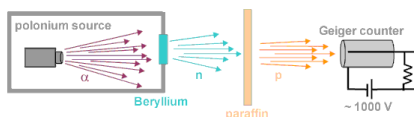
Marking Criteria	Marks
• Both the proton-proton chain and the CNO cycle are outlined with some detail provided • The role of step-wise two particle collisions is emphasised	4
• Both the proton-proton chain and the CNO cycle are outlined in general • The role of two-particle collisions is identified	3
• Both the proton-proton chain and the CNO cycle are identified	2
• Either the proton-proton chain or the CNO cycle is identified	1

The two pathways in which the nucleosynthesis of helium is achieved in stars are known as the proton-proton chain and the CNO cycle. The proton-proton chain begins with a collision between two protons (H nuclei) to form a deuterium nucleus. Another two-particle collision between the deuterium (hydrogen-2) and another proton produces a helium-3 nucleus. Two of these particles may collide to form a helium-4 nucleus.

The CNO cycle also only ever involves the collision of 2 particles in a step-wise process commencing with a carbon nucleus, transmutating in steps through nitrogen and oxygen, ending with the carbon nucleus and the production of a helium nucleus.

Marking Criteria	Marks
- The discovery of the neutron is described clearly in a logical and sequential manner - The neutron is identified as the particle responsible for the existence of isotopes - The nature of the neutron, and how this makes it harder to detect, is clearly linked to why it was discovered later	5
- The discovery of the neutron is described clearly - The neutron is identified as the particle responsible for the existence of isotopes - The nature of the neutron, and how this makes it harder to detect, is linked to why it was discovered later	4
- The discovery of the neutron is outlined - The neutron is identified as the particle responsible for the existence of isotopes	2-3
The neutron is identified as the particle responsible for the existence of isotopes	1

The neutron, a neutral particle with a mass nearly equal that of the proton, was more difficult to discover because of its weak interaction with matter. This resulted in it being discovered much later, after Rutherford had discovered the nucleus but not being able to explain the existence of isotopes. James Chadwick discovered the neutron by using a source of alpha particles aimed at a beryllium block. A sheet of paraffin wax placed behind the beryllium was observed to emit protons as these can be detected by a Geiger counter (whereas neutrons cannot). This investigation confirmed the existence of the neutron.



35 a.

Marking Criteria	Marks
- Correct wavelength calculated - Reasoning why it is not detectable is appropriate	3
- Wavelength calculation made with an error - Shortness of wavelength identified	2
Attempt at wavelength calculation evident	1

Use $\lambda = \frac{h}{mv}$

$$\lambda = (6.626 \times 10^{-34}) / [(1.673 \times 10^{-27}) \times (500 \times 10^3)]$$

$$\lambda = 7.9 \times 10^{-13} \text{ m}$$

The wavelength is much smaller than the diameter of an atom and would not be detectable using conventional methods.

35 b.

Marking Criteria	Marks
- Response includes several different pieces of evidence/observations outlined clearly - Relevant formula is referred to - Response is clear, logical and concise	4
- Response includes a clear outline of some relevant evidence/observation - Relevant formula is identified	3
- Some relevant evidence/observation included - Formula is identified	2
Relevant formula identified directly or indirectly	1

As well as being able to give an explanation for Bohr's 1st postulate that electrons can exist in certain "stationary states" without emitting energy and his 3rd postulate that electrons' angular momentum is quantised, de Broglie's matter wave hypothesis was based on observations made by Davisson and Germer's investigation into the diffraction of electrons using a nickel crystal, a phenomenon only possible if the electrons had a wave nature. Further, electrons used in a double-slit experiment resulted in a diffraction pattern being observed. Measurements taken from these observations confirmed de Broglie's formula that $\lambda = \frac{h}{mv}$, i.e. the wavelength of a particle is Planck's constant divided by the momentum of the particle. Only particles with very small momentum, such as electrons which have a very small mass, would exhibit detectable wavelengths.

36

Marking Criteria	Marks
All required equations are shown correctly in order	3
Most required equations are shown	2
A relevant attempt at a nuclear equation is evident	1

Neutron capture by a silicon-28 nucleus: $^{14}_{28}\text{Si} + {}^1_0\text{n} \rightarrow ^{14}_{29}\text{Si}$

Beta decay of Si-29 to phosphorous: $^{14}_{29}\text{Si} \rightarrow ^{15}_{29}\text{P} + {}^0_{-1}\text{e}$